

ABHIJITH SIVAPRASADAN

Propulsion Test Engineer | High-Temperature Test Rigs | Fluid/Thermal Systems & Instrumentation
Sweden | Open to relocate to Kiruna, Sweden | +46 76 969 2014 | abhijithsivaprasadan@gmail.com |
linkedin.com/in/abhijith-sivaprasadan | github.com/abhijith-sivaprasadan

PROFILE

Mechanical and thermal-fluid engineer completing an M.Sc. in Sustainable Energy Engineering at KTH, with hands-on experience in high-temperature test-rig validation, pressure/temperature instrumentation, NI-DAQ/LabVIEW measurement-chain commissioning, CFD/CHT modelling and Python/MATLAB-based engineering analysis. Strong fit for Isar Aerospace propulsion testing: preparing and troubleshooting test systems, linking models to physical measurements, documenting procedures and validation limits, and working with fluid/thermal hardware under demanding operating conditions. Motivated to apply this foundation to liquid rocket engine development and verification test campaigns in Kiruna.

CORE FIT FOR ISAR AEROSPACE

Development and verification testing: High-temperature test-rig exposure, preliminary test campaign support, instrumentation checks, test-data interpretation, validation planning, troubleshooting mindset and structured technical reporting.

Fluid and thermal systems: Fluid dynamics, compressible flow, thermodynamics, heat transfer, pressure losses, thermal delivery, high-temperature hardware and component-level thermal-fluid evaluation.

Test bench and instrumentation: NI-DAQ, NI MAX, LabVIEW acquisition workflows, thermocouples, static/dynamic pressure sensors, wiring checks, signal verification, measurement-chain commissioning and calibration/verification mindset.

Design-to-realisation mindset: ANSYS Fluent CFD/CHT, Siemens NX, SolidWorks, SpaceClaim, Python/MATLAB analysis, root-cause analysis, documentation of assumptions/limitations and hands-on mechanical systems experience from SAE projects.

EXPERIENCE

Test Engineer / Master Thesis Student

Siemens Energy AB, Fluid Dynamic Lab, Finspång

Apr 2025 - Nov 2025

- Developed steady-state compressible CFD and conjugate heat-transfer models in ANSYS Fluent for high-temperature gas-turbine test-rig hardware, defining boundary conditions, KPIs and evaluation cases.
- Commissioned and verified NI-DAQ measurement chains involving thermocouples, static/dynamic pressure sensors, wiring checks and LabVIEW acquisition workflows for preliminary high-temperature rig testing.
- Conducted mesh/near-wall validation, sensitivity checks and simulation-to-test reasoning to connect model outputs with measurable rig quantities and practical design recommendations.
- Supported experimental troubleshooting through structured documentation of modelling assumptions, validation limits, measurement links and next-step verification needs; contributed to root-cause analysis of heater failure and approved scope adjustment.

Energy Efficiency Intern

Alleima AB, Sandviken (Remote)

Dec 2024 - Jun 2025

- Built an ISO 50001/EED-aligned energy performance mapping approach with KPI/EnPI definitions, baselines, load drivers and normalisation logic for comparing operating states.
- Assessed metering and data readiness; proposed measurement plans for reliable energy follow-up, anomaly detection and improvement tracking in industrial systems.

Backend Engineer

QIBurst, Thiruvananthapuram, India

Aug 2021 - Jul 2023

- Developed and maintained backend APIs, data workflows and automated test pipelines using Python, TypeScript, Postman, Git and CI-oriented development practices.
- Built engineering habits useful for test-bench software and data workflows: structured debugging, reproducibility, version control, test discipline, maintainable delivery and documentation.

Student Intern - Pyrolysis Research

KTH Royal Institute of Technology, Stockholm

Feb 2024 - Jun 2024

- Conducted cost-analysis research and reviewed plastic-pyrolysis reactor concepts to support technology assessment and early-stage process evaluation.

Industrial Intern - Automation, Hydraulics and Pneumatics

Bosch Rexroth India

Jun 2018

- Gained early exposure to industrial automation, hydraulic/pneumatic components, actuator concepts and practical electromechanical system applications.

RELEVANT PROJECTS

SAE Vehicle Design Competitions

SAEINDIA Collegiate Club / Mechanical systems / Team leadership

2019 - 2021

- Led and contributed to BAJA SAE, Bicycle Design Competition and Tractor Design Competition work involving drivetrain/transmission research, mechanical design decisions, specification analysis and technical design reports.
- Coordinated student engineering tasks, vendor/part research, documentation and design reviews, strengthening hands-on mechanical-system thinking and builder mindset.

Turbine Test-Rig CFD, Heat Transfer and Validation

KTH Master Thesis at Siemens Energy / ANSYS Fluent / NI-DAQ

2025

- Built compressible flow and CHT models, performed mesh-independence and near-wall checks, evaluated pressure-loss and thermal-delivery KPIs and translated results into rig-hardware improvement recommendations.
- Used Siemens NX/Teamcenter environment, ANSYS SpaceClaim/Meshing, Python post-processing and LabVIEW/NI-DAQ measurement-chain workflows.

Thermal Systems and Test-Data Analysis Projects

KTH / Python / Energy systems

2024 - 2025

- Applied Python-based modelling and analysis to heating dispatch optimisation, building heating-demand forecasting and thermal energy storage control, strengthening quantitative problem formulation and test-data interpretation skills.

EDUCATION

M.Sc. in Sustainable Energy Engineering (expected 2026)

KTH Royal Institute of Technology, Stockholm

2023 - 2026

- Relevant coursework: Numerical Heat Transfer, Applied Heat and Power Technology, Sustainable Power Generation, Methods of Research in Sustainable Energy, AI Applications in Sustainable Energy Engineering, Energy Systems Modelling.

B.Tech in Mechanical Engineering

APJ Abdul Kalam Technological University, Kerala, India

2017 - 2021

- Relevant foundation: thermodynamics, heat and mass transfer, fluid mechanics, thermal engineering, compressible fluid flow, mechanical systems, machine design, manufacturing and CAD/CAE.

TECHNICAL SKILLS

Fluid / thermal systems:	Fluid dynamics, compressible flow, thermodynamics, heat transfer, thermal systems, pressure loss, high-temperature hardware, gas-turbine test-rig context.
Test / instrumentation:	NI-DAQ, NI MAX, LabVIEW, thermocouples, static/dynamic pressure sensors, wiring checks, signal verification, measurement-chain commissioning.
Test operations:	Test preparation, troubleshooting mindset, procedure documentation, validation planning, safe operations awareness, high-stress and hands-on engineering environments.
Simulation / CAD:	ANSYS Fluent/CHT, ANSYS Meshing, SpaceClaim, Siemens NX, SolidWorks, CFD post-processing, sensitivity studies, mesh/near-wall validation.
Programming:	Python, NumPy, pandas, SciPy, matplotlib, MATLAB scripting basics, TypeScript, Git, Docker, Postman, automated testing.
Engineering delivery:	Technical reports, root-cause analysis, assumptions/limitations tracking, change-request mindset, cross-functional collaboration, SAE mechanical-system leadership.

ADDITIONAL INFORMATION

Certifications: SSG Entre / Industrial Safety | Applied CFD - Siemens | IBM Python for Data Science | Digital Manufacturing & Design Technology | Engineering and Product Design Processes

Leadership: Founder of SAEINDIA Collegiate Club chapter; SAE team lead/captain and secretary/treasurer roles; Students' Nobel NightCap operations and safety role.

Languages: English - fluent/professional | Malayalam - native | Swedish - beginner, actively studying